Chapter 5. Ratio Estimation

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1 Auxiliary Information

So far $y_i = \mu + \varepsilon_i$, $\mathrm{E}(\varepsilon_i) = 0$. With an auxiliary x related to y on each sampled unit, a better model is

$$y_i = f(x_i) + \varepsilon_i, \qquad \mathrm{E}(y_i) = f(x_i) \quad (\text{e.g. } f(x) = \beta_0 + \beta_1 x),$$

and a known μ_x or τ_x sharpens the estimate of μ_y or τ_y: a line through the origin here (ratio estimation), a general line in Ch. 7 (regression).

Example 5.1: Sugar in a truckload of oranges

Orange prices are based on the total sugar content τ_y of a load; $N\bar{y}$ needs the unknown number of oranges N. Sugar y is closely related to weight x, and the load's total weight τ_x is easy to get. Since

$$\frac{\mu_y}{\mu_x} = \frac{N\mu_y}{N\mu_x} = \frac{\tau_y}{\tau_x} \quad \Longrightarrow \quad \tau_y = \frac{\mu_y}{\mu_x}\tau_x,$$

the **ratio estimator** of the total is

$$\hat{\tau}_y = \frac{\bar{y}}{\bar{x}}\tau_x = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i}\tau_x,$$

which does not require N.

Other ratios: food budget to income; children to residents; dose of a new drug relative to a standard one.

2 Ratio Estimation under SRS

Motivation. For x strongly correlated with y and known τ_x, $\hat{R}\tau_x$ is far more stable than $N\bar{y}$.

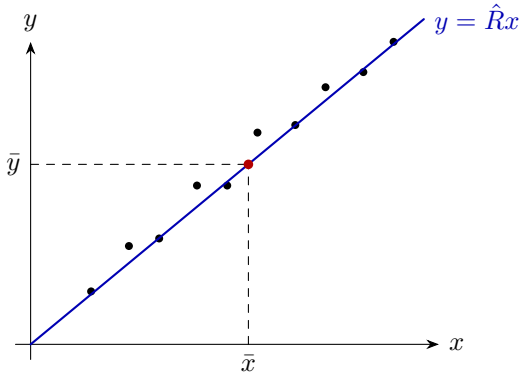


Figure. *A line through the origin with slope $\hat{R} = \bar{y}/\bar{x}$; for $y \approx$ proportional to x, $\hat{\tau}_y = \hat{R}\,\tau_x$ beats $N\bar{y}$.*

Let (x_i, y_i), $i = 1, \dots, n$, be an SRS of pairs from a population of size N. The population ratio and its estimator are

$$R = \frac{\sum_{i=1}^{N} y_i}{\sum_{i=1}^{N} x_i} = \frac{\mu_y}{\mu_x}, \qquad r = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i} = \frac{\bar{y}}{\bar{x}}.$$

Theorem 5.2: Ratio estimators

With $f = n/N$ and $s_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - r x_i)^2$,

ratio:	$r = \bar{y}/\bar{x}$,	$\widehat{\mathrm{Var}}(r) = \dfrac{1-f}{n} \cdot \dfrac{1}{\mu_x^2}\, s_r^2$	(μ_x replaced by $\bar{x}$ if unknown);
total:	$\hat{\tau}_y = r\tau_x$,	$\widehat{\mathrm{Var}}(\hat{\tau}_y) = \tau_x^2 \widehat{\mathrm{Var}}(r) = \tau_x^2 \dfrac{1-f}{n} \dfrac{s_r^2}{\bar{x}^2}$;	
mean:	$\hat{\mu}_y = r\mu_x$,	$\widehat{\mathrm{Var}}(\hat{\mu}_y) = \mu_x^2 \widehat{\mathrm{Var}}(r) = \dfrac{1-f}{n} s_r^2$.	

The residuals $y_i - r x_i$ measure departure from a line through the origin; if $y_i \approx r x_i$, s_r^2 is small.

Derivation: first-order variance of r

Write $d_i = y_i - R x_i$. For N population units, $\bar{D} = \frac{1}{N}\sum_{i=1}^{N}(y_i - R x_i) = \mu_y - R\mu_x = 0$. If $\bar{x} \approx \mu_x$,

$$r - R = \frac{\bar{y} - R\bar{x}}{\bar{x}} \approx \frac{\bar{y} - R\bar{x}}{\mu_x} = \frac{1}{\mu_x} \cdot \frac{1}{n}\sum_{i=1}^{n}(y_i - R x_i) = \frac{\bar{d}}{\mu_x}.$$

Thus $r - R$ is approximately the SRS mean of the d_i divided by μ_x, and by Ch. 2

$$\mathrm{Var}(r) \approx \mathrm{E}(r-R)^2 = \frac{1}{\mu_x^2}\mathrm{E}(\bar{d} - \bar{D})^2 = \frac{1}{\mu_x^2} \cdot \frac{N-n}{N} \cdot \frac{1}{n} S_r^2, \qquad S_r^2 = \frac{1}{N-1}\sum_{i=1}^{N}(y_i - R x_i)^2.$$

Expanding the square,

$$\sum_{i=1}^{N}(y_i - R x_i)^2 = \sum_{i=1}^{N}\left[(y_i - \mu_y) - R(x_i - \mu_x)\right]^2 = (N-1)(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

using $\mu_y = R\mu_x$ and $\sum(x_i - \mu_x)(y_i - \mu_y) = (N-1)\rho S_x S_y$. Hence

$$\mathrm{Var}(r) \approx \frac{1}{\mu_x^2} \cdot \frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y).$$

Replacing R by r and S_r^2 by s_r^2 gives $\widehat{\mathrm{Var}}(r)$. Note that $\mathrm{Var}(r)$ is smallest when $y_i = R x_i$ exactly.

2.1 Correlation form

Expanding s_r^2 gives the equivalent estimate

$$\widehat{\mathrm{Var}}(r) = \frac{1-f}{n} \cdot \frac{1}{\bar{x}^2}(s_y^2 + r^2 s_x^2 - 2r\hat{\rho} s_x s_y), \qquad \hat{\rho} = \frac{s_{xy}}{s_x s_y},$$

with $s_{xy} = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$. A larger positive correlation $\hat{\rho}$ means a smaller variance.

Example 5.3: Assessed values of homes

To estimate the relative change in the assessed value of homes over two years (x: earlier value, y: current value), $n = 20$ homes give

$$r = \frac{\sum_{i=1}^{20} y_i}{\sum_{i=1}^{20} x_i} = 1.066, \qquad \sqrt{\widehat{\mathrm{Var}}(r)} = 0.00745, \qquad B = 2\sqrt{\widehat{\mathrm{Var}}(r)} = 0.0149,$$

so an approximate 95% interval for R is [1.0511, 1.0809].

Example 5.4: Oranges, continued

Ten oranges give $\sum y_i = 0.246$ (sugar) and $\sum x_i = 4.350$ (weight); the load weighs $\tau_x = 1800$ pounds:

$$\hat{\tau}_y = \frac{0.246}{4.350}(1800) = 101.79, \qquad 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau}_y)} = 6.3.$$

3 Bias of the Ratio Estimator

Motivation. The ratio estimator has bias of order $1/n$, negligible relative to its standard error.

r is a ratio of random variables and is generally biased for R. Write $\delta_y = (\bar{y} - \mu_y)/\mu_y$ and $\delta_x = (\bar{x} - \mu_x)/\mu_x$, so

$$r = R\,(1+\delta_y)(1+\delta_x)^{-1} = R(1+\delta_y)(1 - \delta_x + \delta_x^2 - \cdots) \approx R[1 + \delta_y - \delta_x + \delta_x^2 - \delta_x\delta_y].$$

- **Case 1** ($\bar{x} \approx \mu_x$, keep only δ_y): $r \approx R + R\delta_y$, so $\mathrm{E}(r) \approx R$.
- **Case 2** (first order): $\mathrm{E}(r) \approx R + R[\mathrm{E}\delta_y - \mathrm{E}\delta_x] = R$.
- **Case 3** (second order):

$$\mathrm{E}(r) - R \approx R\,\mathrm{E}(\delta_x^2) - R\,\mathrm{E}(\delta_x\delta_y).$$

Theorem 5.5: Approximate bias

$$\mathrm{E}(r) - R \approx \frac{1}{\mu_x^2} \cdot \frac{1-f}{n}(RS_x^2 - \rho S_x S_y).$$

The bias vanishes (to this order) when $RS_x = \rho S_y$, i.e. when $R = \rho S_y/S_x = S_{xy}/S_x^2$ is the least-squares slope.

Proof. Under SRS, $\mathrm{E}[(\bar{x}-\mu_x)^2] = \frac{1-f}{n}S_x^2$ and $\mathrm{E}[(\bar{x}-\mu_x)(\bar{y}-\mu_y)] = \frac{1-f}{n}S_{xy} = \frac{1-f}{n}\rho S_x S_y$. Therefore

$$\mathrm{E}(r) - R \approx \frac{R}{\mu_x^2}\frac{1-f}{n}S_x^2 - \frac{R}{\mu_x\mu_y}\frac{1-f}{n}\rho S_x S_y,$$

and $R/(\mu_x\mu_y) = 1/\mu_x^2$. ∎

Remark 5.6: When is r unbiased?

If $y_i = \beta x_i$ (a line through the origin), then $R = \sum y_i / \sum x_i = \beta$ and also

$$\rho \frac{S_y}{S_x} = \frac{S_{xy}}{S_x^2} = \frac{\frac{1}{N-1}\sum (x_i - \mu_x)\,\beta(x_i - \mu_x)}{\frac{1}{N-1}\sum (x_i - \mu_x)^2} = \beta,$$

so the bias term is zero. The bias is negligible whenever the relation between y and x is a straight line through the origin.

Theorem 5.7: An exact bias bound

$$\mathrm{E}(r) - R = -\frac{\mathrm{Cov}(r, \bar{x})}{\mu_x}, \qquad \frac{|\mathrm{E}(r) - R|}{\sqrt{\mathrm{Var}(r)}} \le \frac{\sqrt{\mathrm{Var}(\bar{x})}}{\mu_x} = CV(\bar{x}).$$

Proof. Since $r\bar{x} = \bar{y}$,

$$\mathrm{Cov}(r, \bar{x}) = \mathrm{E}(r\bar{x}) - \mathrm{E}(r)\mathrm{E}(\bar{x}) = \mu_y - \mathrm{E}(r)\mu_x = \mu_x(R - \mathrm{E}(r)).$$

By $|\mathrm{Cov}(r, \bar{x})| \le \sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$, $|\mathrm{E}(r) - R| \le \frac{1}{\mu_x}\sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$. ■

The bias is small relative to the SE when the CV of $\bar{x}$ is small (moderately large n).

4 Ratio Estimation in Stratified Random Sampling

Motivation. Ratios per stratum (separate) or overall (combined), depending on stratum sizes and ratio differences.

With strata $h = 1, \ldots, L$ (weights $w_h = N_h/N$), there are two ways to build a ratio estimator of μ_y.

Definition 5.8

1. **Separate ratio estimator**: estimate the ratio within each stratum and combine,

$$\hat{\mu}_{yRS} = \sum_{h=1}^{L} w_h\, r_h\, \mu_{xh}, \qquad r_h = \frac{\bar{y}_h}{\bar{x}_h}.$$

   For two strata A and B: $\hat{\mu}_{yRS} = \frac{N_A}{N}\frac{\bar{y}_A}{\bar{x}_A}\mu_{xA} + \frac{N_B}{N}\frac{\bar{y}_B}{\bar{x}_B}\mu_{xB}$.

2. **Combined ratio estimator**: estimate μ_y and μ_x by the stratified means and take one ratio,

$$\hat{\mu}_{yRC} = r_C\, \mu_x, \qquad r_C = \frac{\bar{y}_{\mathrm{st}}}{\bar{x}_{\mathrm{st}}}, \quad \bar{y}_{\mathrm{st}} = \sum_h w_h \bar{y}_h, \ \bar{x}_{\mathrm{st}} = \sum_h w_h \bar{x}_h.$$

Theorem 5.9: Estimated variances

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yRS}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h} \cdot \frac{\sum_{i=1}^{n_h} (y_{hi} - r_h x_{hi})^2}{n_h - 1},$$

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yRC}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h} s_{rh}^2,$$

where s_{rh}^2 is the sample variance of the terms $y_{hi} - r_C x_{hi}$ in stratum h.

Derivation: which one is better?

The population versions are $\mathrm{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,S}^2$ and $\mathrm{Var}(\hat{\mu}_{yRC}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,C}^2$, with $R_h = \mu_{yh}/\mu_{xh}$, $R_C = \mu_y/\mu_x$ and

$$\begin{aligned}(N_h-1)S_{h,S}^2 &= \sum_i (y_{hi} - R_h x_{hi})^2 = (N_h-1)(S_{yh}^2 + R_h^2 S_{xh}^2 - 2R_h\rho_h S_{xh}S_{yh}),\\ (N_h-1)S_{h,C}^2 &= \sum_i \left[(y_{hi}-\mu_{yh}) - R_C(x_{hi}-\mu_{xh})\right]^2\\ &= (N_h-1)(S_{yh}^2 + R_C^2 S_{xh}^2 - 2R_C\rho_h S_{xh}S_{yh}).\end{aligned}$$

(The combined deviation $\hat{\mu}_{yRC} - \mu_y \approx \bar{y}_{\mathrm{st}} - R_C\bar{x}_{\mathrm{st}} = \bar{d}_{\mathrm{st}}$ is a stratified mean of $d_{hi} = y_{hi} - R_C x_{hi}$, whose stratum means are $\mu_{yh} - R_C\mu_{xh}$.) Subtracting,

$$\begin{aligned}S_{h,C}^2 - S_{h,S}^2 &= (R_C^2 - R_h^2)S_{xh}^2 - 2(R_C - R_h)\rho_h S_{xh}S_{yh}\\ &= (R_C-R_h)^2 S_{xh}^2 + 2(R_C-R_h)(R_h S_{xh}^2 - \rho_h S_{xh}S_{yh}).\end{aligned}$$

The last bracket is zero when each stratum ratio is (approximately) unbiased, $R_h S_{xh} = \rho_h S_{yh}$. Then

$$\mathrm{Var}(\hat{\mu}_{yRC}) - \mathrm{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h}(R_C - R_h)^2 S_{xh}^2 \;\geq 0.$$

Remark 5.10: Practical guidance

- When the stratum ratios R_h differ greatly, the separate estimator is more efficient.
- But the stratum ratio biases add up: badly biased for small n_h, negligible for n_h over about 50. With small strata use the combined estimator.

5 Ratio Estimator versus the Sample Mean

Motivation. Ratio beats $\bar{y}$ when ρ exceeds half the ratio of the coefficients of variation.

Compare $\bar{y}$ and $\hat{\mu}_y = r\mu_x$ by relative efficiency $\mathrm{RE}(\hat{\theta}_1, \hat{\theta}_2) = \mathrm{Var}(\hat{\theta}_1)/\mathrm{Var}(\hat{\theta}_2)$:

$$\mathrm{Var}(\bar{y}) = \frac{1-f}{n}S_y^2, \qquad \mathrm{Var}(\hat{\mu}_y) \approx \frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

$$\mathrm{RE}(\bar{y}, \hat{\mu}_y) = \frac{\mathrm{Var}(\bar{y})}{\mathrm{Var}(\hat{\mu}_y)} = \frac{S_y^2}{S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y}.$$

Theorem 5.11: When the ratio estimator wins

$\mathrm{RE}(\bar{y}, \hat{\mu}_y) > 1$ if and only if

$$\rho > \frac{1}{2}\cdot\frac{S_x/\mu_x}{S_y/\mu_y} = \frac{1}{2}\cdot\frac{CV_x}{CV_y}.$$

In particular, if x and y have equal coefficients of variation, the ratio estimator is better as soon as $\rho > 1/2$.

Proof. $\mathrm{RE} > 1 \iff R^2 S_x^2 < 2R\rho S_x S_y \iff RS_x < 2\rho S_y \iff \rho > \frac{RS_x}{2S_y}$, and $R = \mu_y/\mu_x$. The sample version replaces everything by estimates: $\hat{\rho} > \frac{1}{2}\frac{s_x/\bar{x}}{s_y/\bar{y}}$. ∎

Efficient for highly positively correlated x, y; the estimated relative efficiency is meaningful only for large samples.

6 Selected exercises

Motivation. The ratio estimator is model-optimal, and a computation for a total.

Exercise 5.1 (The ratio estimator as a weighted least-squares fit)

Consider the model $y_i = \beta x_i + \varepsilon_i$, $i = 1, \dots, n$, with fixed $x_i > 0$, $\mathrm{E}(\varepsilon_i) = 0$, $\mathrm{Var}(\varepsilon_i) = \sigma^2 x_i$ and uncorrelated errors.
(a) Show that $r = \sum y_i / \sum x_i$ minimizes the weighted sum of squares $\sum_i (y_i - \beta x_i)^2 / x_i$.
(b) Show that r is unbiased for β with $\mathrm{Var}(r) = \sigma^2 / \sum_i x_i$, and that it has the smallest variance among all linear unbiased estimators $\sum_i a_i y_i$ of β.

Solution. [Supplement]

(a) $Q(\beta) = \sum_i (y_i - \beta x_i)^2 / x_i$ is a quadratic in β with $Q'(\beta) = -2\sum_i (y_i - \beta x_i) = -2(\sum_i y_i - \beta \sum_i x_i)$ and $Q'' = 2\sum_i x_i > 0$, so the unique minimizer is $\beta = \sum y_i / \sum x_i = r$.

(b)

Step 1. (Unbiasedness and variance.) $r = \sum_i a_i^0 y_i$ with $a_i^0 = 1/\sum_j x_j$. Then $\mathrm{E}(r) = \beta \sum_i a_i^0 x_i = \beta$ and $\mathrm{Var}(r) = \sigma^2 \sum_i (a_i^0)^2 x_i = \sigma^2 / \sum_i x_i$.

Step 2. (Unbiasedness constraint.) A linear estimator $\sum_i a_i y_i$ has mean $\beta \sum_i a_i x_i$, so it is unbiased for every β iff $\sum_i a_i x_i = 1$. Its variance is $\sigma^2 \sum_i a_i^2 x_i$.

Step 3. (Cauchy–Schwarz.) Writing $a_i x_i = (a_i \sqrt{x_i})\sqrt{x_i}$,

$$1 = \Big(\sum_i a_i x_i\Big)^2 \le \Big(\sum_i a_i^2 x_i\Big)\Big(\sum_i x_i\Big),$$

so $\sigma^2 \sum_i a_i^2 x_i \ge \sigma^2 / \sum_i x_i = \mathrm{Var}(r)$, with equality iff $a_i \sqrt{x_i} \propto \sqrt{x_i}$, i.e. a_i constant, i.e. $a_i = a_i^0$.
So it is BLUE when $y \propto x$ with error variance $\propto x$, as for the oranges. ■

Exercise 5.2 (Sales of television sets)

Total TV sales τ_y this quarter: an SRS of $n = 13$ of $N = 123$ offices gives previous-year x_i and current y_i, with previous-year total $\tau_x = 128{,}200$.

x_i	550	720	1500	1020	620	980	928	1200	1350	1750	670	729	1530
y_i	610	780	1600	1030	600	1050	977	1440	1570	2210	980	865	1710

Estimate τ_y by the ratio estimator and place a bound on the error.

Solution.

Step 1. $\sum x_i = 13{,}547$, $\sum y_i = 15{,}422$, so $r = 15{,}422/13{,}547 = 1.1384$ and $\hat\tau_y = r\tau_x = 1.1384 \times 128{,}200 = 145{,}944$.

Step 2. The residual variance is $s_r^2 = \frac{1}{12}\sum_i (y_i - r x_i)^2 = 12{,}990$, and $\mu_x = \tau_x / N = 1042.3$ is known.

Step 3. By the ratio-estimator theorem,

$$\widehat{\mathrm{Var}}(\hat\tau_y) = \tau_x^2 \frac{1-f}{n} \frac{s_r^2}{\mu_x^2} = N^2 \frac{1-f}{n} s_r^2 = 123^2 \cdot \frac{1 - 13/123}{13} \cdot 12{,}990 = 1.352 \times 10^7,$$

so the bound is $2\sqrt{1.352 \times 10^7} = 7354$ and τ_y is estimated by $145{,}944 \pm 7354$.

Step 4. For comparison, $N\bar y = 123 \times 1186.3 = 145{,}916$ has $s_y^2 = 231{,}543$ and bound

$$2N\sqrt{(1-f)s_y^2/n} \approx 31{,}050 :$$

the ratio estimator is about four times as precise, since current and previous sales are almost proportional. ■